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Assessing the impact of a safety income model as an alternative approach to universal basic income: a case study in South Korea

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Abstract

Purpose – This case study focuses on an alternative minimum-security model, the Safety Income (SI) Model, representing a nuanced departure from both established policies and UBI. StI is a welfare system that supplements households earning below the standard median income with 50% of the difference between the standard median and their current earnings. The quantitative case study presents the set-up of SI and assesses the cost of its implementation in South Korea. By employing a computable general equilibrium model method, the study compares the impacts of SI, UBI and the existing scheme in South Korea on income disparities, labor market outcomes and Gross Domestic Product.

Design/methodology/approach – In the past decade, the Universal Basic Income (UBI) concept has gained international significant traction as a potential remedy for poverty and inequality. However, the practical implications of UBI implementation remain under extensive debate. It is unclear if UBI is an effective model for poverty alleviation.

Findings – The analyses show that SI outperforms the other two welfare systems across all studied economic indicators. SI demonstrates more substantial reductions in income inequality compared with UBI and the existing scheme, minimal impact on unemployment rates compared with other schemes and a relatively modest decrease in GDP, making it a more favorable choice for South Korea when developing the minimum-security system within the specified budget constraint.

Originality/value – This research contributes to the discourse surrounding basic income, economic security, poverty alleviation and inclusive social policies.

Keywords Poverty alleviation, Universal basic income, Safety income, Basic social security, South Korea

Paper type Case study

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Introduction

Over the past few years, the idea of a Universal Basic Income (UBI) has surfaced as a groundbreaking solution to tackle urgent challenges associated with global poverty and inequality (Van Parijs and Vanderborght, 2019). The idea of providing an unconditional, regular income to all citizens, regardless of income level or employment status, has gained traction in both Global North and Global South context. The surge of interest has been further amplified by the COVID-19 pandemic experience, prompting governments worldwide to reevaluate social safety nets (De Wispelaere and Morales, 2021; Mäntyneva and Hiilamo, 2024). While the theoretical promise of UBI is tantalizing, the practical implications of its implementation remain a subject of ongoing debate. As of now, no country has adopted a full UBI model, leaving critical questions about its efficacy, affordability and overall impact unanswered (e.g. Kangas *et al.*, 2021). This gap in understanding underscores the need for comprehensive studies that compare UBI with alternative models designed to enhance economic security and well-being. This study focuses on the Safety Income (SI) Model, a nuanced approach that represents a departure from both the existing policy framework and the theoretical construct of UBI (Park, 2016).

Basic income proponents, who are mostly coming from political philosophy background (e.g. Van Parijs and Vanderborght, 2019), argue that UBI could significantly reduce financial poverty. As Chrisp (2020) argues although implementing basic income in its ideal form may not be politically viable in the short-term, it has the potential to greatly influence welfare state politics by rallying political actors and voters to support substantial reforms to the social security system. However, empirical analyses demonstrate that poverty-reducing impact of BI hinges on nuanced implementation choices concerning, for example the level and specifications of BI, coupled with decisions on maintaining, modifying or replacing elements of the existing tax-benefit system (Aerts *et al.*, 2023). Simulations question whether a UBI is fiscally realistic in high-income countries (Green *et al.*, 2023; Hoynes and Rothstein, 2019). This study contributes to the debate on economic impacts of UBI in the context of South Korea as well as to global debate on the feasibility of UBI.

The background for the SI model in South Korea relates to an incident in 2014, when the Korean society was shocked by the suicides of a mother in her 60s and her two adult daughters in their 30s in Songpa-gu, a district in Seoul. The assumed cause of the suicide was that they could not afford their livelihood expenses. While the South Korean government has enforced the National Basic Livelihood Security System (NBLSS) since 2000, it has been consistently criticized for creating a large welfare blind zone where many remain unprotected (Chae, 2022; Choi and Kim, 2022). Suicides of families in extreme economic circumstances have consistently occurred, despite the government's significant effort to expand the NBLSS. What is worse, they have even increased since the burst of the COVID-19 pandemic. In November 2022, Korean society witnessed again the suicides of a mother in her 60s and two daughters in their 30's in Suwon city, south of Seoul, a *deja-vu* of the 2014 tragedy (Kwon, 2022). After the 2014 tragedy, while the total welfare budget in South Korea increased by 61.0% from 151.7 billion in 2015 to 244.2 billion US dollars in 2021, the proportion of households receiving cash benefits from NBLSS increased by only 5.2% 2015 to 7.4% during the same period (Korean Statistical Information Service, 2022).

Many people are now questioning whether the current NBLSS has accomplished or will ever accomplish its goal of effectively abolishing the welfare blind zone. Reflecting this widespread discontent, some politicians and scholars proposed alternatives. Until now, Universal Basic Income (UBI) and Safety Income (SI) are the most frequently discussed alternative welfare systems in South Korea. UBI is an older and more widely known alternative to existing welfare systems in South Korea. UBI was proposed as one of the top pledges during the 2022 presidential campaign in South Korea by the presidential candidate of the Democratic Party, who then now stepped down to the opposition. Gyeonggi Province

has piloted youth basic income with positive evaluations (Yoo *et al.*, 2019). SI, proposed by Park (2016), is newer approach which is gaining attraction as well as practical applications. SI was among the top promises of the incumbent Mayor of Seoul, who was elected in April 2021 and reelected in June 2022. Adopting SI's philosophy and welfare policy implications, Seoul Metropolitan Government launched the pilot project of SI (SSIP) in 2022 that provides cash assistance to selected households earning an income below a minimum threshold for three years (Seoul Metropolitan Government, 2022) [1]. Choosing between SI, UBI and the expansion of the current cash welfare benefits of NBLSS has emerged as one of the hottest political and social policy issues in South Korea since 2021.

Through a detailed examination of this alternative, this study seeks to illuminate the potential benefits, challenges and comparative advantages it offers, particularly in the unique socio-economic landscape of South Korea. By juxtaposing SI against both a reform in established policy schemes (NBLSS) and against UBI, the study aims to provide insights into the potential implications of adopting SI model. Moreover, by setting up a computable general equilibrium model and utilizing it for the comparison of SI, UBI and NBLSS, the study analyzes the effects of three alternatives on three economic indicators: income differentials, labor market outcomes and Gross Domestic Product. The research aims to answer questions: firstly, how much would it cost to replace existing policy scheme in South Korea (NBLSS) with SI model, and secondly, if that amount of money would be spend on revamping NBLSS or to introduce UBI how they would compare with SI in terms on income differentials, labor market outcomes and Gross Domestic Product. With the Korean case study, the study aims to contribute to the broader discourse surrounding economic security, poverty alleviation and inclusive social policies.

The following section II of the article discusses the minimum social security context in Korea and the set-up of the SI model. Section III focuses on theoretical backgrounds and policy issues surrounding SI. Section IV estimates the cost needed for the implementation of SI in the South Korean context. The next section V compares SI's economic effects with UBI and the expansion of NBLSS. The final section VI presents conclusions on the feasibility of the SI model as opposed to UBI or expansions of NBLSS.

Existing policy context: NBLSS and EITC

The NBLSS in South Korea is the basic support program for low-income households. It comprises seven complementary benefits: livelihood, housing, education, medical, self-reliance, childbirth and funeral benefits. The livelihood benefit is a means-tested cash transfer for a four-member household with a recognized income of 22,590 US dollars ($) [2] or less in 2022 [3]. The amount of cash transfer is the gap between this amount and the recognized income [4].

The housing benefit is also a cash transfer for a rental household of a recognized income of $34,640 or less in 2022. The education benefit is also a means-tested cash transfer for tuition, textbook expenses and educational activity support. The medical benefit covers almost all medical or pharmaceutical expenses of the household members with a recognized income of $30,120 or less in 2022. Under the self-reliance benefit the government provides the household members with public-paying jobs to support themselves [5]. The earned income tax credit (EITC) of the National Tax Service (NTS) is also a cash transfer for a household with a market income of $36,760 or less (e.g. for the single earner in the household) and a maximum net asset of $245,100 [6]. Additionally, there is the child tax credit of NTS.

A commonly acknowledged problem is that the multiple means-tested benefits reduce work incentives. Lack of incentive to work has been consistently pointed out as one of the major problems of the current welfare system. Those receiving EITC support in 2018 accounted for only 14.8% of all NBLSS beneficiaries who were able to work (Park *et al.*, 2021,

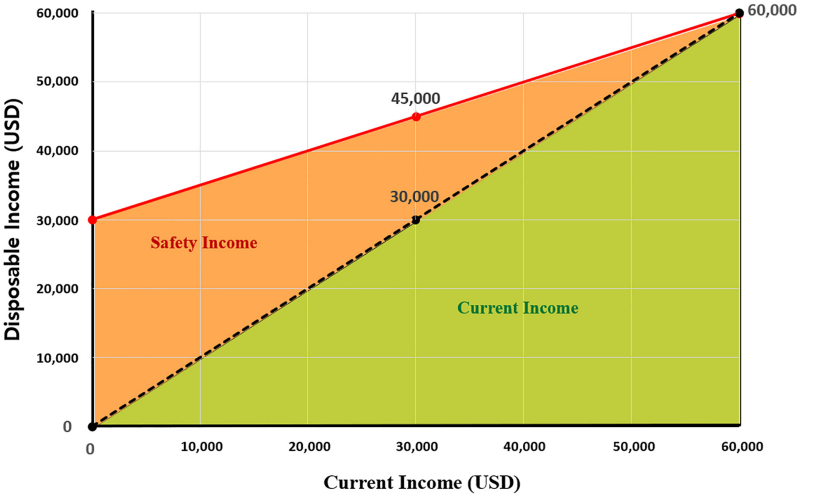
p. 94). Another issue is low coverage. Theoretically, approximately 15% of all households should be able to receive the livelihood benefit, but only 4.6% receive it. One major reason for this discrepancy is the government's imposing various strict restrictions on eligibility for support. The restrictions include the beneficiaries' income, assets, automobiles and family dependents [7].

The set-up of the safety income model

Park (2016, 2017, 2020) together with Park and Byun (2017) proposed that all households in Korea earning less than the standard median income receive SI support, 50% of the difference between the standard median income and their current income. However, households earning more than the standard median income should not receive SI.

Figure 1 shows the basic structure of SI by demonstrating the pattern that SI decreases as household income increases. If it is assumed that the standard median income of four-member households is $60,000, then four-member households with no income receive $30,000 (50% of the median income $60,000), while households earning $60,000 receive none. In the middle, households earning $30,000 receive $15,000 [50% of ($60,000 – $30,000)]. While livelihood, housing, self-reliance benefits and earned income and child tax credits will be integrated into SI, NBLSS's four other benefits are maintained. In addition to NBLSS, other national welfare benefits such as basic pension, national pension, unemployment benefits and child allowance are also maintained. Since the amount of SI decreases as household income increases, SI is said to follow the principle of "thick bottom, thin top."

The SI is similar to the Negative Income Tax (Friedman, 1962, p. XII) in that the two provide a household of its income lower than the standard level with the payment of 50% of the difference between the standard level and its income. The SI, however, is quite different from NIT in the following three respects. First, while NIT abolishes all existing welfare benefits, SI maintains almost all of them. Second, while NIT was suggested with a flat tax rate on income by Friedman (1962, p. X), SI maintains the current tax system with many forms of deductions. Third, while NIT suggested by Friedman covers only the lowest 10–33%, SI covers as much as the lowest 50% of households.



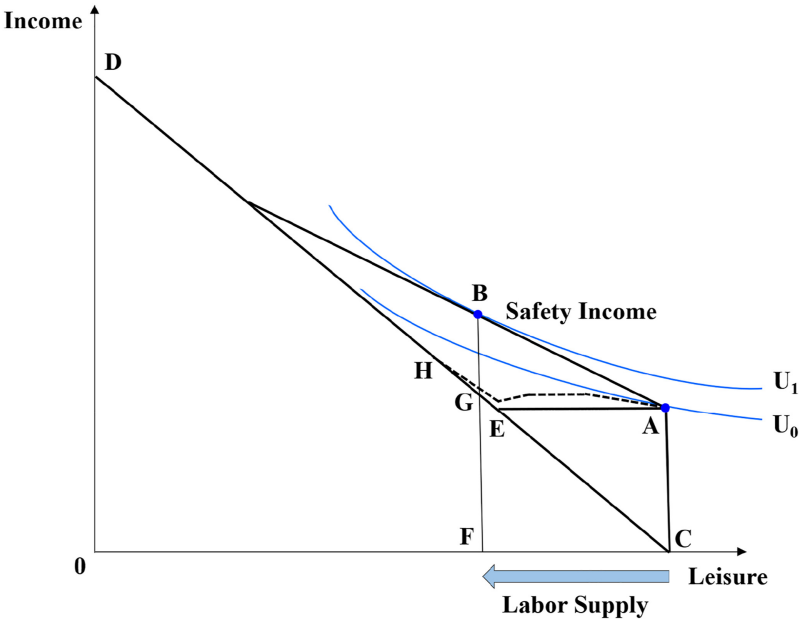
Source(s): Authors own work

Figure 1.
The basic structure of safety income

Another feature of SI is that the amount of SI is determined only by the income of each household and "paid in advance and adjusted afterward" just as income tax is withheld monthly and settled at the end of the year, following the principle of "support forward, settle afterward." The idea is that SI would prevent economic hardship and their tragic consequences more effectively than the current NBLSS.

Yet another feature of SI is that it would improve work incentives compared with existing safety net [8]. Figure 2 shows that labor-leisure choice changes by receiving SI in 2019. In Figure 2, the line CD is the wage line: The labor income increases with working hours, and the slope shows the wage rate. The line CAE indicates the livelihood and housing benefits, while the dotted line A to H indicates EITC giving out different credits for different income intervals. With the existing welfare system of NBLSS if household members can work, they will choose point A, where they receive the maximum livelihood and housing benefits. Since the reservation wage rate exceeds 37% of their market wage rate, at which EITC starts, they are better off staying on A rather than working. If so, there is in theory no labor supply with the current welfare system of NBLSS. If livelihood and housing benefits are replaced by SI, as indicated by the red line in Figure 2, the households' choice moves from A to B. Now, there is a labor supply with the new welfare system SI. When this occurs, its disposable income increases (BF > AC), its utility increases ($U_1 > U_0$) and the amount of government payment to the household decreases (BG < AC). For details of SI see Supplementary file (Number of households and average benefit amounts by benefit category).

In short, the SI model would cover more households in terms of both income and wealth compared to the NBLSS benefits (or NIT suggested by Friedman). SI would include greater benefits than NBLSS. SI would also decrease at a slower rate than NBLSS and thereby increase incentives for work.



Source(s): Authors own work

Figure 2.
The change of labor-leisure choice by safety income model

Estimating the additional budget for enforcing SI

In the following text the practical implications for implementing SI model are evaluated first by estimating arithmetically the additional budget for enforcing SI. The estimation is needed to assess the macro-level implications which are analyzed in the second section. In the macro-level analysis SI is contrasted with the alternatives where similar amount of money would be spent on UBI or extension of current minimum income schemes.

Following the method adopted by Park (2016) the total cost for enforcing SI is defined as the current income minus the public transfer income such as livelihood and unemployment benefits. The additional budget for SI is estimated as follows: First, 100% of the budget for "livelihood, housing, self-reliance benefits, and earned income and child tax credits" is subtracted from the estimated total cost. Second, 50% of the budget for the other public transfer incomes is subtracted from it.

The total cost for SI is estimated to be $88,620m in 2019 with Korea Welfare Panel (KWP) data. The "livelihood, housing, self-reliance benefits, and earned income and child tax credits" are substituted with SI support. To calculate the additional budged for SI the annual budget for these items, $12,786m, is subtracted from the estimated total cost.

The 2018 cash social expenditure of the government of $94,960m [9] includes the expenditure for "livelihood, housing, self-reliance benefits, and earned income and child tax credits." If SI were to be enforced, the market income and the expenditure for all the other public transfers, where the expenditure of all the other public transfers is calculated as $82,174m ($94,960m - $12,786m), would add up to the current household income. Therefore, 50% of $82,174m, which is $41,087m, would also save the budget for enforcing SI. Consequently, the additional budget for SI is estimated as:

$88,620m − $12,786m − 0.5 × ($94,960m − $12,786m) = $34,747m. Since the 2019 cash social expenditure of the government increased from that of 2018, the additional budget for SI would be estimated to be less than $34,747m in 2019.

In SI, all households below the standard median income can receive SI regardless of the amount of assets they own. In the above formula, $82,174m should be paid to households below the standard median income, but due to incongruences in implementation a certain portion of the amount is also paid to households with earnings more than the standard median income. This increases the additional budget for SI by that much. The estimated additional budget for enforcing SI is subject to change due to the yearly budget structure and income distribution of the beneficiaries.

The budget for welfare, labor and health service has increased annually by 11% between 2017 and 2022 (Ministry of Economy and Finance, 2017, 2022). Suppose the central government's budget growth rate for welfare, labor and health service between 2017 and 2022 maintains until 2027. In that case, the 2027 budget will increase by $174,182m from the 2022 budget. The estimated additional budget for SI, $34,747m, would be just 19.9% of the increment. The disaster subsidy paid seven times since the break of the COVID-19 pandemic has been $116,589m in total. Therefore, the estimated additional budget for SI would be 29.8% of the disaster subsidy, much smaller than the recent subsidy in June of 2022, $45,561m.

While the SI benefit amounts are greater for lower-income households, the benefits of UBI are uniformly small for all households regardless of their income level. If an estimated additional budget of $34,747m to enforce SI were evenly distributed to all people as UBI, each person would receive $710 per year, and four-member households would receive $2,840 ($710 × 4) per year. In contrast, in the SI scheme, four-member households with no income would receive $32,340 per year, 11.4 times the UBI [10]. So, assuming the same budget constraint, UBI would practically become a "thin bottom, thin top" welfare system, and SI, a "thick bottom, thin top" welfare system.

Macro-level effects of SI, UBI and expansion of NBLSS

SI, UBI and the expansion of the current cash welfare benefits (NBLSS would change recipients economic behavior including consumption, savings and labor supply, to maximize their utility subject to the changed budget constraint. If, for instance, all citizens receive the significant amount of support as in UBI, their consumption and savings may increase and their labor supply may decrease. Consequently, the aggregate consumption, investment, employment, output, etc. would change and so their market income and budget constraint would also change. Their utility maximization subject to the newly changed budget constraint would also change the aggregate variables, and the above process would repeat infinitely. To provide an estimate of the outcomes, it is possible to utilize a general equilibrium model (Pissarides, 1990), in which the above process ultimately would reach an equilibrium. A general equilibrium model is a theoretical framework used to analyze the interactions and interdependencies among various economic factors in an entire economic system. The model aims to capture the simultaneous interactions of supply and demand in multiple markets, considering how changes in one market can affect others.

In this study the general equilibrium model is used to analyze and compare certain macro-level effects of SI, UBI and the expansion of NBLSS in South Korea. In the employed computable general equilibrium model, households maximize their utility with their current income after paying taxes and being paid public transfers, and firms hire workers, utilize machines and facilities (capital) and produce output to maximize their profit. The government collects taxes and makes expenditures on various sectors. Using a computable general equilibrium model, an analysis is conducted to evaluate the effects of SI, UBI and the expansion of NBLSS on three outcomes which have been in center of debate concerning new minimum-security models and especially UBI (Hiilamo, 2022). They are income differentials, employment and GDP (see the assumptions and the details of the model in Appendix 2).

Impact on income differentials

This study assesses the potential of social security models to mitigate income differential by two measures, Gini coefficient and income quintile ratio (distance between richest and poorest quintile). Assuming the additional budget of $34,747m is spent on the enforcement of SI, it would have two effects, direct and indirect, on the economy. The direct effect comes from the income transfer from the government to households. The indirect effect, contrarily, comes from the changes in the economic behavior of households and firms, including their consumption, investment and labor supply. As shown in Table 1 (results shown by household income deciles), the direct effect of SI decreases the Gini coefficient by 6.91% to 0.361 and the income quintile ratio by 24.84% to 7.02.

Meanwhile the direct effect of UBI reduces the Gini coefficient by 1.36% to 0.383 and the income quintile ratio by 4.69% to 8.9. Since UBI increases the transfer income of all households including high-income households by the same amount, its effect on mitigating the income gaps among households is much smaller than SI.

The direct effect of the expansion of NBLSS reduces the Gini coefficient by 2.26% to 0.379 and the income quintile ratio by 5.51% to8.83. Since the welfare benefits are given regardless of households' income, they would increase the transfer income of high-income households, and its effect on mitigating the income gaps among households would be smaller than SI. Since the current NBLSS is concentrated on low-income households, the expansion of NBLSS would narrow the income gaps among households more than UBI.

The indirect effect of SI would also reduce the Gini coefficient by 0.10% or 0.00039 unit but increase the income quintile ratio by 0.12% or 0.01104 units. As will be seen in the next section, if SI were enforced, the numbers of the employed would increase in 8, 9 and 10 deciles of household income so that the income quintile ratio would increase. The total effect of SI

Table 1.
The effects of three welfare systems on mitigating household income differentials

Income differential index	Welfare system	Criterion coefficient (a)	Direct effects		Indirect effects		Final effects	
			Coefficient (b)	Change from criterion (%) (b/a-1) x100	Coefficient (c)	Change from criterion (%) (c/a-1) x100	Coefficient (d)	Change from criterion (%) (d/a-1) x100
Gini Coefficient	SI	0.38826	0.36142	−6.91289	−0.00039	−0.10045	0.36103	−7.01334
	UBI	0.38826	0.38297	−1.36249	0.00056	0.14423	0.38353	−1.21826
	NBLSS	0.38826	0.37947	−2.26395	0.00025	0.06439	0.37972	−2.19956
Income Quintile Ratio	SI	9.34129	7.02113	−24.83768	0.01104	0.11818	7.03234	−24.71768
	UBI	9.34129	8.90287	−4.69336	0.09688	1.03712	8.99975	−3.65624
	NBLSS	9.34129	8.82644	−5.51155	0.09058	0.96967	8.91702	−4.54188

Source(s): Authors own work

would reduce the Gini coefficient by 7.01% to 0.3610 and the income quintile ratio by 24.72% to 7.03.

UBI and the expansion of NBLSS also have indirect effects by changing the behavior of economic agents such as consumption, investment and labor supply. First, the indirect effect of UBI increases the Gini coefficient by 0.14% or 0.00056 units and the income quintile ratio by 1.04% or 0.09688 unit. The total effect of UBI would reduce the Gini coefficient by 1.22% to 0.3835 and the income quintile ratio by 3.66% to 9.0. Second, the indirect effect of the expansion of NBLSS increases the Gini coefficient by 0.06% or 0.00025 units and the income quintile ratio by 0.97% or 0.09058 units. The total effect of the expansion of NBLSS reduces the Gini coefficient by 2.20% to 0.3797 and the income quintile ratio by 4.54% to 8.92.

Effects on the labor markets and GDP

Since the work incentive and the labor supply decrease as the transfer income from the government to consumers increases by $34,747m, the number of people employed would decrease generally. Table 2 shows the changes in the number of employed persons, unemployed persons and economically active persons (employed persons + unemployed persons) by household income bracket. While both UBI and the expansion of NBLSS would decrease the number of employed in all deciles, SI would increase the number of employed in 8, 9 and 10 deciles. In total, UBI and NBLSS would decrease the employed by 219 thousand and 277 thousand, respectively, but SI, by 186 thousand. UBI would increase the unemployed by 84 thousand but SI only by 9 thousand. The expansion of NBLSS would make 1,063 thousand unemployed drop out the labor market so that the labor force would decrease by 1,340 thousand.

Table 3 shows the changes in unemployment rate and percentage change from current (2019) rate by household income bracket. The SI would decrease the unemployment rates in 1, 2, 7 and 8 deciles and increase the total unemployment rate only by 0.03% points. On the other hand, UBI and NBLSS would increase the unemployment rates in all deciles and the total unemployment rates by 0.30% points. The results show that SI outperforms UBI and NBLSS in terms of reducing the number of employed people the least and increasing the number of unemployed and the unemployment rate the least.

The general equilibrium produces direct estimates for the GDP change. While UBI and the expansion of NBLSS would decrease GDP by 0.54% and 0.49%, SI decreases GDP only by 0.24%. The reason why SI decreases GDP is because the income effect of $34,747m of the public transfer is more than offsetting the labor supply effect of SI. However, that effect is smaller than for UBI or expansion of NBLSS.

Conclusion

This is the first study to evaluate the effects of safety income model (SI). SI is welfare system that aims to effectively eliminate welfare blind zones and alleviates poverty in South Korea. By providing support to only those who fall short of a certain income level and more to those who need it more, SI is designed to guarantee a certain standard of living while maintaining incentives for work. Previous literature on innovative social security models has almost exclusively focused on UBI (e.g. Hiilamo, 2022). Moreover, there have been only very few studies analyzing the macro-level effects of basic income or other transformative social security models (e.g. OECD, 2017).

The additional budget needed to enforce SI in 2019 was estimated to be $34,747m (see Appendix 3 for number of households and average benefit amounts by benefit category). If that amount would be used for UBI, I, each person would receive $710 per year and 4-member households would receive $2,840 ($710 × 4) per year, while in the SI scheme, 4-member

Table 2.
The effects of three welfare systems on the change of employment, unemployment and economically active population by household income bracket (unit: Number of persons)

Household Income bracket	SI			UBI			NBLSS		
	Employed	Unemployed	Economically active	Employed	Unemployed	Economically active	Employed	Unemployed	Economically active
W01	−12,849	−11,193	−24,042	−22,624	5,848	−16,776	−25,864	5,520	−20,344
W02	−53,620	−2,811	−56,430	−25,155	7,808	−17,347	−34,097	7,248	−26,849
W03	−67,705	8,951	−58,754	−26,264	9,549	−16,715	−41,085	9,882	−31,203
W04	−34,437	5,155	−29,282	−24,761	9,996	−14,765	−42,030	11,864	−30,166
W05	−29,125	6,808	−22,316	−27,414	13,664	−13,750	−53,691	19,244	−34,448
W06	−20,462	4,295	−16,167	−27,355	11,876	−15,479	−31,942	10,690	−21,252
W07	−2,051	−2,360	−4,411	−24,225	9,480	−14,744	−25,386	6,993	−18,392
W08	6,053	−573	5,480	−16,837	4,284	−12,553	−13,326	3,088	−10,238
W09	6,197	300	6,497	−14,517	5,011	−9,506	−7,719	3,489	−4,230
W10	21,902	−13	21,889	−9,548	6,503	−3,045	−2,277	5,380	3,103
Total	−186,096	8,559	−177,537	−218,701	84,020	−134,681	−277,416	83,396	−194,020

Source(s): Authors own work

households with no income would receive $32,340 per year, 11.4 times the UBI . The results demonstrate how UBI would practically become a "thin bottom, thin top" welfare system, while SI is a "thick bottom, thin top" welfare system.

By setting up a computable general equilibrium model, the analysis compared the effects of SI, UBI and the expansion of the current NBLSS on the economy, assuming the same budget constraint of $34,747m spent on each welfare system. The results demonstrate the SI outperformed over the other two welfare systems as follows. First, SI reduces income gaps significantly more than UBI and NBLSS. For example, while UBI and NBLSS reduce the Gini-coefficient only by 1.2% and 2.2%, SI reduces it by 7.0%. Second, SI increases the unemployment rate only by 0.03% percentage points, while UBI and NBLSS increase it by 0.3% percentage points. Third, SI decreases GDP only by 0.24%, while UBI and NBLSS decrease it by 0.54% and 0.49%, respectively. Notably, SI demonstrates more substantial reductions in income inequality, minimal impact on unemployment rates compared with other schemes and a relatively modest decrease in GDP, making it a favorable choice within the specified budget constraint.

Based on the analysis above SI could eliminate at least some of the blind zone of the current welfare system and support beneficiaries much more effectively than UBI or expanding the current NBLSS. SI has an absolute advantage in alleviating poverty over UBI in that it provides "thick" benefits to those who really need it by following the "thick bottom, thin top" principle. On the contrary, UBI almost certainly ends up with "thin bottom, thin top" if both welfare systems are subject to the same budget constraint. In that sense SI would meet the financial needs of welfare recipients to maintain a decent life that UBI cannot guarantee due to its budget constraint.

Aligning with the results from Aerts *et al.* (2023), Green *et al.*, 2023 and Hoynes and Rothstein (2019) the findings question the effectiveness of UBI in reducing income inequalities in high-income countries. The findings also emphasize the need to look for other novel social security models beyond UBI to tackle existing challenges. They may, as demonstrated in this study, outperform UBI in terms of realizing favorable economic outcomes.

Everyone in any society could fall into a trap in economic life. It is possible to compare the welfare systems to insurance, both SI and UBI resemble auto insurance. Just as auto insurance allows the insured to drive safely, SI ensures that everyone conducts economic activities safely. Auto insurance is paid to the insured according to the extent of an accident, only in the event of an accident. Similarly, SI is paid to only those who fall short of a certain income level and more to those who need it more, following the principle of "thick bottom, thin

Household income bracket	Current unemployment rate (%)	SI		UBI		NBLSS	
		Unemployed (%)	Change (%p)	Unemployed (%)	Change (%p)	Unemployed (%)	Change (%p)
W01	7.25	5.85	−1.4	7.98	0.73	7.94	0.69
W02	4.34	4.16	−0.18	4.85	0.50	4.81	0.47
W03	5.23	5.64	0.4	5.66	0.43	5.68	0.45
W04	4.78	4.97	0.2	5.16	0.38	5.23	0.45
W05	9.4	9.61	0.21	9.82	0.43	10.00	0.60
W06	4.69	4.82	0.13	5.05	0.36	5.02	0.32
W07	3.03	2.97	−0.07	3.31	0.27	3.23	0.20
W08	0.80	0.78	−0.02	0.92	0.12	0.89	0.09
W09	1.07	1.08	0.01	1.21	0.14	1.16	0.10
W10	1.81	1.81	0.00	1.98	0.17	1.95	0.14
Total	3.77	3.80	0.03	4.07	0.30	4.07	0.30

Source(s): Authors own work

Table 3. The effects of three welfare systems on unemployment rate and percentage change from current rate (2019) by household income bracket

top." However, UBI fundamentally differs from SI. UBI is likened to a malfunctioning auto insurance, which pays the same small amount to all insured regardless of whether or not an accident occurs or the extent of an accident.

A limitation of this study is that it was based on the use of computable general equilibrium model. It is a theoretical framework which has clear limitations. For example, it assumes perfect competition, rationality and homogeneous agents. Most importantly the model cannot estimate the type of dynamic effects and large behavioral responses suggested by UBI advocates. The limitations do not render the model useless, but they highlight the importance of recognizing its simplifying assumptions and considering the real-world complexities that may not be captured in the model. The evaluation of the ongoing pilot project on a version of SI (SSIP) in the city of Seoul will provide empirical evidence on the impacts of SI in real life.

A limitation of SI system is that requires efficient data systems. Realizing the "support forward, settle afterward" principle necessitates that the amount of SI is determined only by the income of each household and paid in advance and adjusted later. In the Korean context, the National Tax Service (NTS) in Korea has the most comprehensive and precise information on the incomes and assets of citizens among all governmental institutions. Similar institutions operate in other developed countries. Future research should compare SI's effects in South Korea and elsewhere on other outcomes besides income, for example on mental health, life and job satisfaction and social relationships.

Notes

1. The SSIP differs from Park's (2016) original SI as follows. First, households below 85% of the standard median income and below the net asset value of $399,510 are eligible for SSIP. Second, while livelihood, housing benefits and basic pension are incorporated into SSIP, self-reliance benefits, earned income tax credit and child tax credit are maintained. In addition, Seoul's own welfare programs, such as Seoul Basic Security Benefit and Seoul Youth Allowance, are incorporated. The key differences between SI and SSIP are summarized in Table A1 in Appendix 1.
2. This amount is 8,436,000 wons (₩), 30% of the standard median income, ₩61,453,000 ($75,310) in 2022. Based on the 2020 median income, the standard median income in 2022 was determined to be $75,310 by the Ministry of Health and Welfare. The study uses the purchasing power parity exchange rate, ₩816 per US dollar in 2022, citing the OECD Statistics. The dollar denomination of all Korean wons is rounded from 10 digits in the text.
3. The households used as examples in this paper are all assumed to be 4-member households.
4. The converted income has been 70% of labor or business income and 100% of other incomes since 2020. Before 2020, it was 100% of labor or business income and other incomes. Each property, including the car, has its conversion rate into income. The recognized income of a household is the summation of all converted incomes.
5. The childbirth and funeral benefits are for households whose recognized income is lower than $34,640 in 2022. The childbirth benefit is to give $860 per newborn baby, and the funeral benefit is to give $980 per dead person. While the housing and education benefits are administered by the Ministry of Land, Infrastructure and Transport and the Ministry of Education, respectively, the other benefits are by the Ministry of Health and Welfare.
6. In detail, the amount of credit is 37.1% of the household's labor or business income lower than $8,580, $3,190 for the household whose labor or business income is between $8,580 and $17,280 and decreases with the household's labor or business income which is between $17,280 and $36,760 to zero for the household whose labor or business income exceeds $36,760.
7. Owning an automobile is probably the most notorious restriction of all. A household with a car with a resale value of $3,000. Since the car's value is converted to an annual income of $36,000 and exceeds 30% of the standard median income, this household cannot receive the livelihood benefit at all.

8. A household member without the market income who receives SI of $37,650, 50% of the standard median income in 2022 would be willing to work when he or she is given a job offer that guarantees annual earnings of $32,280. If he or she works at the job, his or her disposable income is $32,280 + 0.5 × ($75,310 - $32,280) = $53,800: Since the disposable income increases by $16,150, he or she is less likely not to work.
9. This is the latest available number. The data source is from OECD Statistics.
10. According to the 2019 KWP data, the maximum livelihood benefit is $19,010 for a 4-member household, and the maximum housing benefit, $2,080. When a household receives both benefits, the total benefit is $21,090 per year. Therefore, if SI were enforced, a household's disposable income would increase by at least $11,250 ($32,340 - $21,090) to $32,340. By doing so, SI will considerably contribute to narrowing the income gaps among households.

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Appendix

The Appendix for this article can be found online.

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